

Supplementary information:

One bucket record measures both scene and medium

[author block — USER DECISION (mirrors main text, gap G7)]

This supplement is the full scientific record behind the main article: proofs, the DLGI model and estimator, the demoted dead-time sibling (ridge law and M1 campaign), the five preregistered closure certificates, the ruthless prior-art fence, the confirmatory protocol, failure maps, and provenance. Section, equation, table, and figure numbers carry an “S” prefix; “main-text Sec. X” refers to the primary manuscript. Internal method acronyms and audit labels appear only in Notes S6 and S8 (R36 s2, s6).

Notation

[Include the reconciled notation table from `../NOTATION.tex` (gap G1); collisions are flagged there. When assembling, strip its `documentclass` wrapper and include only the `longtable` body.]

- S1 Regularity conditions and the joint missing-information partition**
- S2 Canonical-confusion reciprocity: proof and determinant identities**
- S3 DLGI model, gauge, identifiability, estimator, and calibration**
- S4 Dead-time sibling: hidden-state ridge law and the M1 campaign**
- S5 Auxiliary hidden-state and schedule derivations**
 - S5.1 Detector-side escape beyond the count-only model (retired)**
- S6 Five preregistered closure certificates**
- S7 Prior-art fence**
- S8 Confirmatory protocol and the C1–C7 campaign**
- S9 Edge maps, residual and model checks, and sensitivity**
- S10 Provenance, correction record, and reproduction**